

HFSS SIMULATION REPORT

5.65 GHz Printed / Flexible Antenna Design

Ansys Electronics Desktop 2025 R2.4 – HFSS

1. Objective

The objective of this work was to design and simulate a compact printed/flexible antenna targeted around 5.65 GHz using Ansys HFSS. The model includes a copper radiating structure, feed line, ground structure, dielectric substrates, coaxial feed, lumped port and an open radiation region. The antenna was evaluated using the input reflection coefficient S11.

2. Software and Simulation Setup

Parameter	Value
Software	Ansys Electronics Desktop 2025 R2.4
Solver	HFSS Driven Modal / Modal Network
Design frequency	5.65 GHz
Frequency sweep	5.3–6.0 GHz; final response examined over 4–8 GHz
Excitation	Lumped Port
Port impedance	50 Ω
Outer boundary	Radiation boundary
Main output	dB(S(1,1)) / S11

3. Model Components and Materials

Component	Material	Purpose
PCB substrate	FR4 epoxy	Dielectric support
FPC substrate	Polyimide	Flexible dielectric layer
Radiator / trace	Copper	Radiating conductor
Feed line	Copper	RF feed
Coax dielectric	PTFE	Coax insulation
Coax pin	Conductor	Center conductor
Coax shield	Conductor	Return path
Airbox	Vacuum / air	Free-space radiation region

4. Main Design Parameters

Variable	Value
f0	5.65 GHz
fmin	5.3 GHz
fmax	6.0 GHz
PCB L × W	15 × 10 mm
PCB thickness	0.8 mm
Copper thickness	0.035 mm
FPC L × W	23 × 10 mm
FPC thickness	0.1 mm
Arm length	11.5 mm
Arm width	1 mm

Feed length	4 mm
Feed width	1.2 mm
Ground pad L × W	8 × 8 mm
Coax length	15 mm
Coax center conductor diameter	≈0.3 mm
Coax dielectric diameter	≈0.95 mm
Coax shield diameter	≈1.15 mm
Air gap	25 mm

5. HFSS Modeling Procedure

1. Define the design variables and target frequency.
2. Create the FR4 and polyimide dielectric bodies and assign their materials.
3. Create the copper radiator, arm and feed-line geometry.
4. Create the ground structure and feed clearance.
5. Model the coaxial feed using the conductive pin, PTFE dielectric and shield.
6. Create the surrounding air/vacuum region.
7. Assign the required conductor boundaries and verify the electrical connections.
8. Create a 50 Ω lumped port at the feed and define its integration line.
9. Assign the radiation boundary to the outer air region.
10. Create Setup1 at 5.65 GHz and define the frequency sweep.
11. Validate the design, solve the setup and generate the S11 report.

6. Excitation and Boundary Conditions

A lumped port with 50 Ω reference impedance was used to excite the antenna. The port represents the RF source at the feed region. The outer airbox was assigned a radiation boundary to approximate an open/free-space environment. The integration line of the lumped port defines the voltage direction and must lie correctly on the port plane.

7. Simulation Result

The final HFSS simulation completed successfully and the frequency sweep converged. The principal result is dB(S(1,1)), representing the input reflection coefficient.

Result	Observed value
Target frequency	5.65 GHz
Main simulated resonance	≈ 6.0 GHz
Minimum S11	≈ -34 dB
Port impedance	50 Ω
Simulation status	Sweep converged / normal completion

8. Discussion

The S11 curve shows a clear resonance close to 6 GHz. The minimum of approximately -34 dB indicates strong simulated impedance matching at that frequency. Since the nominal design frequency is 5.65 GHz, the antenna can be further tuned if an exact 5.65 GHz resonance is required. Radiator/arm length, feed position, feed dimensions and ground dimensions are key parameters for frequency tuning and impedance matching.

During development, HFSS reported an overlap warning between Copper_Arm1 and Copper_Feed_Line. This should be reviewed in a final geometry revision so that the conductor geometry and boundary assignments accurately represent the intended physical structure.

9. Conclusion

A parameterized 5.65 GHz printed/flexible antenna model was successfully developed and simulated in Ansys HFSS. The final S11 simulation converged and produced a pronounced resonance near 6.0 GHz with a minimum S11 of approximately -34 dB. The result demonstrates good simulated impedance matching near resonance and provides a baseline for further optimization toward the desired 5.65 GHz operating point.

10. Future Work

- Tune the radiator/arm length to shift resonance toward exactly 5.65 GHz.
- Perform parametric sweeps of arm length, feed position and ground dimensions.
- Calculate -10 dB impedance bandwidth and VSWR.
- Generate Smith Chart and input impedance plots.
- Generate 2D/3D radiation patterns and evaluate gain, directivity and efficiency.
- Resolve/verify any remaining conductor-overlap warnings before final reporting.

End of Report